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Azure Cosmos DB troubleshooting documentation

Welcome to Azure Cosmos DB troubleshooting. These articles explain how to determine, diagnose, and fix issues that you might experience when you use Azure Cosmos DB. In the navigation pane, browse through the article list or use the search to find specific issues and solutions.

General troubleshooting

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Troubleshoot Azure Cosmos DB Emulator

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Data Explorer fails to connect

Troubleshoot the Azure Cosmos DB emulator

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The Azure Cosmos DB emulator provides an environment that emulates the cloud service for development. Use the tips in this article to help troubleshoot problems that you might experience when you install or use the emulator.

Troubleshooting checklist

Here's a list of common troubleshooting steps to follow if the Azure Cosmos DB emulator isn't working as expected.

Reset data

If you installed a new version of the emulator, and you're experiencing errors, make sure that you reset the data. To reset the data, open the Azure Cosmos DB emulator context menu from the system tray, and then select **Reset Data**.

If resetting the data doesn't fix the errors, you can:

- Uninstall the emulator.
- Uninstall older versions of the emulator (if they exist).
- Remove the `%ProgramFiles%\Azure Cosmos DB Emulator` folder.
- Reinstall the emulator.

Alternatively, if resetting the data doesn't work, go to the `%LOCALAPPDATA%\CosmosDBEmulator` location, and then delete the folder.

Review corrupted Windows performance counters

- If the Azure Cosmos DB emulator stops responding, collect the dump files from the `%LOCALAPPDATA%\CrashDumps` folder, compress the files, and then open a support ticket in the Azure portal.

- If `Microsoft.Azure.Cosmos.ComputeServiceStartupEntryPoint.exe` stops responding, this crash could indicate that the performance counters are corrupted. To check the counter status, run the following command:

  Windows Command Prompt

```
lodctr /R
```

Diagnose connectivity problems

- If you experience a connectivity problem, collect trace files, compress the files, and then open a support ticket in the Azure portal.

- If you receive a "Service Unavailable" message, the emulator might not be initializing the network stack. Check to see whether you have the **Pulse Secure Client** or **Juniper Networks Client** installed because their network filter drivers might be causing the problem. You can also try uninstalling other network filter drivers to fix the problem. Alternatively, start the emulator by using `/DisableRIO` to switch the emulator network communication to regular Winsock.

- If you receive a connectivity error message such as `"Forbidden", "message":"Request is being made with a forbidden encryption in transit protocol or cipher. Check account SSL/TLS minimum allowed protocol setting..."`, this error message might indicate global changes in the OS (for example Insider Preview Build 20170) or changes in the browser settings that enable TLS 1.3 as the default protocol. A similar error message, such as "Microsoft.Azure.Documents.DocumentClientException: Request is being made with a forbidden encryption in transit protocol or cipher. Check account SSL/TLS minimum allowed protocol setting" might be generated if you use the SDK to run a request against the Azure Cosmos DB emulator. This error might also occur because the Azure Cosmos DB emulator supports only the TLS 1.2 protocol. The recommended workaround is to set TLS 1.2 as the default.

  For example:

  1. In **IIS Manager**, go to **Sites** > **Default Web Sites**.

  2. Locate the **Site Bindings** for port **8081**, and edit them to disable TLS 1.3. You can also update the settings for the web browser by using the **Settings** option.

     > **Note**
     >
     > If your computer enters sleep mode or runs any OS updates while the emulator is running, you might see a "Service is currently unavailable" error message.

  3. To reset the emulator data, right-click the icon that appears in the Windows notification tray, and then select **Reset Data**.

Collect trace files

To collect debugging traces, run the following commands as an administrator in a Command Prompt window:

1. Navigate to the path in which the emulator is installed:

   Windows Command Prompt

   ```
   cd /d "%ProgramFiles%\Azure Cosmos DB Emulator"
   ```

2. Shut down the emulator, and watch the system tray to make sure that the program is shut down:

   Windows Command Prompt

   ```
   Microsoft.Azure.Cosmos.Emulator.exe /shutdown
   ```

   > **Note**
   >
   > It might take one minute for the process to finish. You can also select **Exit** in the Azure Cosmos DB emulator user interface.

3. Start logging by running the following command:

   Windows Command Prompt

   ```
   Microsoft.Azure.Cosmos.Emulator.exe /startwprtraces
   ```

4. Start the emulator:

   Windows Command Prompt

   ```
   Microsoft.Azure.Cosmos.Emulator.exe
   ```

5. Reproduce the problem. If the data explorer isn't working, you have to wait only a few seconds for the browser to load to be able to detect the error.

6. Stop logging:

   Windows Command Prompt

```
Microsoft.Azure.Cosmos.Emulator.exe /stopwprtraces
```

7. Navigate to the `%ProgramFiles%\Azure Cosmos DB Emulator` path, and locate the *docdbemulator_000001.etl* file.

8. Open a support ticket in the Azure portal, and include the .etl file together with any steps that are required to reproduce the problem.

Install certificates for client applications

Occasionally, you might need to take the exported emulator certificate and use it with a client application. The exact process varies by SDK.

Export TLS/SLL certificate

Export the emulator certificate to successfully use the emulator endpoint from languages and runtime environments that don't integrate with the Windows Certificate Store. You can export the certificate using Windows Certificate Manager or PowerShell after you run the emulator for the first time.

1. Retrieve the certificate using the friendly name `DocumentDbEmulatorCertificate` and store the certificate in a shell variable named `$cert`.

   PowerShell

   ```
   $cert = Get-ChildItem Cert:\LocalMachine\My | where{$_.FriendlyName -eq
   'DocumentDbEmulatorCertificate'}
   ```

2. Use Export-Certificate to export the certificate to a temporary file in your home folder.

   PowerShell

   ```
   $params = @{
       Cert = $cert
       Type = "CERT"
       FilePath = "$home/tmp-cert.cer"
       NoClobber = $true
   }
   Export-Certificate @params
   ```

> **Note**

In Windows, the home folder is typically `C:\Users\[username]\`, assuming your home drive is `C:`.

3. Use certutil to convert the certificate to a **Base-64 encoded X.509** certificate file.

```
PowerShell
certutil -encode $home/tmp-cert.cer $home/cosmosdbcert.cer
```

4. Remove the temporary file.

```
PowerShell
Remove-Item $home/tmp-cert.cer
```

Import certificate for Java applications

When you run Java applications or MongoDB applications that use a Java-based client, installing the certificate into the Java default certificate store is easier than passing the `-Djavax.net.ssl.trustStore=<keystore> -Djavax.net.ssl.trustStorePassword="<password>"` parameters. For example, the included Java Demo application (`https://localhost:8081/_explorer/index.html`) depends on the default certificate store.

Follow the instructions in Creating, Exporting, and Importing TLS/SSL Certificates to import the X.509 certificate into the default Java certificate store. Remember that you're working in the *%JAVA_HOME%* directory when running keytool. After the certificate is imported into the certificate store, clients for SQL and Azure Cosmos DB's API for MongoDB can connect to the Azure Cosmos DB emulator.

Alternatively, you can run the following `bash` script to import the certificate:

```
Bash
#!/bin/bash

# If the emulator was started with /AllowNetworkAccess, replace the following with
the actual IP address of it:
EMULATOR_HOST=localhost
EMULATOR_PORT=8081
EMULATOR_CERT_PATH=/tmp/cosmos_emulator.cert
openssl s_client -connect ${EMULATOR_HOST}:${EMULATOR_PORT} </dev/null | sed -ne
'/-BEGIN CERTIFICATE-/,/-END CERTIFICATE-/p' > $EMULATOR_CERT_PATH
# Delete the cert if it already exists
sudo $JAVA_HOME/bin/keytool -cacerts -delete -alias cosmos_emulator
```

```
# Import the cert
sudo $JAVA_HOME/bin/keytool -cacerts -importcert -alias cosmos_emulator -file
$EMULATOR_CERT_PATH
```

Once the `CosmosDBEmulatorCertificate` TLS/SSL certificate is installed, your application should be able to connect to and use the local Azure Cosmos DB emulator.

If you have any issues, see Debugging SSL/TLS connections. In most cases, the certificate might not be installed into the *%JAVA_HOME%/jre/lib/security/cacerts* store. For example, if there's more than one installed version of Java, your application might be using a certificate store different from the one you updated.

Import certificate for Python applications

When you connect to the emulator from Python applications, TLS verification is disabled. By default, the Python SDK for Azure Cosmos DB for NoSQL doesn't try to use the TLS/SSL certificate when it connects to the local emulator. For more information, see Azure Cosmos DB for NoSQL client library for Python.

If you want to use TLS validation, follow the examples in TLS/SSL wrapper for socket objects.

Import certificate for Node.js applications

When you connect to the emulator from Node.js SDKs, TLS verification is disabled. By default, the Node.js SDK (version 1.10.1 or later) for the API for NoSQL doesn't try to use the TLS/SSL certificate when it connects to the local emulator.

If you want to use TLS validation, follow the examples in the Node.js documentation.

Rotate certificates

You can force regeneration of the emulator certificates by opening the emulator with the `/ResetDataPath` argument. This action wipes out all the data stored locally by the emulator. For more information about command-line arguments, see Windows emulator command-line arguments.

> **Tip**
>
> Alternatively, select **Reset Data** from the Azure Cosmos DB emulator's context menu in the Windows system tray.

If you installed the certificates into the Java certificate store or used them elsewhere, you must reimport them using the current certificates. Your application can't connect to the local emulator until you update the certificates.

Deployment of private endpoints for Azure Cosmos DB fails

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Azure Private Link endpoints can be deployed inline for an Azure Cosmos DB account through an Azure Resource Manager template (ARM template). However, this deployment might fail if particular prerequisites aren't met.

Symptoms

When you try to deploy an ARM template, you receive an error message that states that the `Microsoft.Network/virtualNetworks/write` permission is required.

Cause

The `Microsoft.Network/virtualNetworks/write` permission is required to deploy a private endpoint inline for an Azure Cosmos DB account. This permission isn't shown in the list of required permissions to deploy a private endpoint on its own. For more information, see role-based access control permissions for private endpoints.

This issue occurs only if the private endpoint is deployed inline for the Azure Cosmos DB account.

Solution

Make sure that the deploying principal is granted the `Microsoft.Network/virtualNetworks/write` granular permission before you use an ARM template to deploy an Azure Cosmos DB account that has an inline private endpoint.

Reference

- What is a private endpoint?
- Create an Azure Cosmos DB Account with a private endpoint

Contact us for help

If you have questions or need help, create a support request, or ask Azure community support. You can also submit product feedback to Azure feedback community.

Azure Cosmos DB Data Explorer fails to connect

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Occasionally, the Azure Cosmos DB Data Explorer can't connect to your account or perform operations against resources or items. This article reviews potential causes and solutions to this issue.

> **Important**
>
> The Azure Cosmos DB Data Explorer isn't available for the API for PostgreSQL or the API for vCore.

Prerequisites

An existing Azure Cosmos DB for NoSQL, MongoDB RU, Apache Cassandra, Apache Gremlin, or Table account

Symptoms

You're unable to connect to the Data Explorer even after enabling the **Allow access from Azure Portal** option.

Cause

Even after you configure the correct role-based access control and portal permissions, there are extra network access requirements that must be configured based on your selected API.

- Some services, such as the API for NoSQL, API for Apache Gremlin, and API for Table, use a client-side JavaScript SDK to perform operations and require one set of solutions.
- Other services, such as the API for MongoDB RU and API for Apache Cassandra, use protocol-specific middleware and require an alternative solution.

Solution for the API for NoSQL, Apache Gremlin, or Table

Database, container, graph, and table (control plane) operations are executed via calls to the Azure Resource Manager control plane using the Azure Cosmos DB resource provider. Your network configuration doesn't affect these operations.

Item (data plane) operations are executed using the JavaScript SDK within the context of your browser. The device you're currently using must have direct network access to the account.

- For accounts configured with **public access (all networks)**, data plane operations shouldn't have any network-related connectivity issues.
- For accounts configured with **public access (selected networks)**, you must create a firewall rule to allow access to the account from your current device. The **Networking** feature in the service's page in the Azure portal has an **Add my current IP (...)** hyperlink that can automatically add your IP address.
- If the account doesn't have a firewall rule, the Data Explorer throws an error that can be observed in the notifications for Data Explorer. This error message contains text similar to the following examples:

Expand table

API	Error
NoSQL	`Request originated from IP XXX.XXX.XXX.XXX through public internet. This is blocked by your Cosmos DB account firewall settings.`
Apache Gremlin	`Failure in submitting query: g.V(): Request originated from IP XXX.XXX.XXX.XXX through public internet. This is blocked by your Cosmos DB account firewall settings.`
Table	`Error while refreshing databases: Request originated from IP XXX.XXX.XXX.XXX through public internet. This is blocked by your Cosmos DB account firewall settings.`

- For accounts configured with **public access disabled**, your device must be connected to a virtual network with connectivity to the account for data plane operations to function.

Solution for the API for MongoDB RU or Apache Cassandra

Review the control plane and data plane solutions, respectively.

Database, keyspace, and collection (control plane) operations are executed via calls to the Azure Resource Manager control plane using the Azure Cosmos DB resource provider. Your

network configuration doesn't affect these operations.

Item (data plane) operations are executed using a proxy service in the API's middleware. This service facilitates the use of the service-native (MongoDB, Cassandra) protocol in item operations and queries. The proxy service requires direct network access to the account.

- For accounts configured with **public access (all networks)**, data plane operations shouldn't have any network-related connectivity issues.
- For accounts configured with **public access (selected networks)**, you must select **Allow Access from Azure Portal** in the **Networking** feature of the service's page in the Azure portal. This option adds a range of IP addresses that include the middleware service. You *don't* have to add the IP address of your current device.
- If the account doesn't have a firewall rule, the Data Explorer throws an error that can be observed in the browser developer tools. This error message contains a generic `Error querying documents` or `Failed to establish connection with cassandra node` message generated by the middleware service. The expanded error message is included in the Compute gateway logs, similar to the following example:

  Output

  ```
  Request originated from IP XXX.XXX.XXX.XXX through public internet. This is
  blocked by your Cosmos DB account firewall settings.
  ```